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Psychological research frequently is concerned with latent constructs that cannot be observed directly, such as intelligence or personality. These constructs are inferred from manifest indicators, such as item responses or behavioral measures, using measurement models. The quality of these models is central to substantive inference: if a measure does not adequately represent the construct of interest, even highest statistical rigor cannot prevent misleading conclusions (Flake and Fried 2020; Flake et al. 2017). However, recent work has emphasized the need for stronger statistical methods (Flake et al. 2017; Haig 2020; Altoè et al. 2020) but also more explicit theoretical foundations in psychology (Eronen and Bringmann 2021). We argue that measurment quality is essential to both developments. A key aspect of measurement quality is measurement invariance (MI), that is, the assumption that a measurement model functions equivalently across groups, samples, or time points. So MI ensures that a latent construct is measured in the same way across groups, conditions, or occasions. This assumption is required whenever latent means, variances, covariances, or structural relations are compared, making MI a cornerstone of most correlational and experimental research designs. Violations of MI imply that the relationship between the latent construct and its observed indicators differs across groups or conditions. In such cases, differences in observed responses may not solely reflect differences in the underlying construct but may also arise from systematic differences in item functioning or measurement precision. Consequently, observed group differences become ambiguous, as they may represent a combination of substantive construct differences and measurement-related distortions. Without establishing MI, it is therefore impossible to determine whether apparent differences reflect true variation in the latent construct or artifacts of the measurement process. Consequently, MI is not only a methodological requirement but also a prerequisite for valid theoretical interpretation.

Classical Measurement Invariance Testing

Probably due to the nature of research questions, concerning latent constructs, Structural Equation Models (SEMs) have become a widely used approach to model such relations. In that SEMs represent hypotheses about data containing latent constructs, by defining relations of observable, minfest indicators and suggested latent factors, on so called measurement models. Those models parameters can be estimated by minimizing a suitable discrepancy function, which also provides sample-based measures of model fit. By separating construct-related variance from measurement error, latent variable models provide a more robust basis for inference than approaches that rely exclusively on observed variables.

MI testing in the classical sense is usually done using Multi-group confirmatory factor analysis (MGCFA), sometimes also referred to as Multi-group Structural Equation Modeling (MGSEM). Thereby MI testing usually proceeds through a sequence of multiple-group model comparisons in which equality constraints are imposed on increasingly restrictive sets of parameters, such as factor loadings, intercepts, and residual variances. Equality of the models structure in both groups is called configural invariance, and can be understood as both groups share the models general form (i.e. number of latent factors and items). Equality of factor loadings former is referred to as metric invariance, if metric invariance is present it indicates that all factor loading parameters are equal across groups. Specifically, this suggests that the items of a given scale and their underlying constructs relate to each other with the same strengths in both groups. Furthermore, it is called scalar invariance if the item intercepts are invariant across groups as well. If scalar MI is established, this suggests that in addition to the factorial structure and the factor loadings, the vectors of the item intercepts are also invariant. Scalar invariance is necessary for the comparison of latent mean differences across groups, because it indicates that the scales that were used have the same unit of measurement as well as the same origin. Residual or strict invariance assumes, additionally to the constraints above, that variables residuals are equal across the groups. If the residual variance holds, the items of the scale are measuring the latent constructs with the same measurement error. This implies that it can indeed be known whether differences measured by the items of the tested scale between the groups are due to group differences in the proposed factors or due to measurement errors. What is only implied in those interpretations of the different MI-levels, is that the question which level specifically one "needs" to hold in ones one data is determined by ones research question and the goal of the inference. As mentioned, metric invariance for example is sufficient to warrant latent mean comparisons between groups. In many empirical studies this is what researchers aim to do, compare latent mean or sum scores of individuals of different groups, then metric invariance suffices as MI-level.

Traditional approaches to measurement invariance (MI) testing are largely confirmatory in nature. Researchers begin by specifying a structural equation model (SEM), typically a confirmatory factor analysis (CFA) model, that reflects substantive hypotheses about the measurement structure and relationships among variables. Subsequently, a categorical grouping variable is selected to partition the sample into two or more groups, and equality constraints are imposed and tested across these groups to evaluate different levels of MI. Although this procedure is confirmatory, it nevertheless depends on several researcher decisions. In practice, two choices are particularly consequential. First, the measurement model must be specified, including the number of factors and the pattern of relationships between latent and observed variables. Second, a grouping variable must be selected and operationalized. This choice is typically guided by theoretical expectations regarding potential sources of heterogeneity in the data. Consequently, the detection of measurement non-invariance is restricted to the specific grouping variable under consideration and may overlook heterogeneity arising from other observed or unobserved characteristics.

Moderated Nonlinear Factor Analysis

However, MGCFA testing also has some limitations (e.g.: groups need to be created by categorical variables; the differences of said groups are assumed to be linear in their difference) that were overcome with more contemporary approaches to MI-testing. One example, that is getting more popular, is the Moderated nonlinear factor analysis (MNLFA). By conceptualizing MI differently, it enables the possibility to have the groups being created via continuous and potentially nonlinear variables. In MNLFA testing Measurement non-invariance (MNI) is modeled as the moderation of the measurement model (or parts of it) by a variable, external to that measurement model. That moderator variable is the variable, that was the former “grouping variable” in the MGCFA approach. The understanding is that if any model parameters are moderated by that external variable, then the measurement model is not measurement invariant, because it performs differently dependent on that variable.

The exogenous variable may influence any subset of the parameters in the measurement model. Specifically, the means, variances, and covariance of latent factors, as well as the item intercepts, factor loadings, and residual variances of any manifest indicator, may vary systematically across individuals as deterministic functions of the exogenous variable. Whether MI is present depends on which parameters are moderated by the exogenous variable. If moderation is limited to the latent-factor parameters—that is, the means, variances, and covariance — the model remains consistent with MI. In contrast, moderation of item-level parameters, including intercepts, loadings, or residual variances, constitutes MNI. Accordingly, a moderation function must be specified for each parameter that is allowed to vary as a function of the exogenous variable. Because of this conceptualization, the exogenous variable, the moderator, can be categorical as well as continuous, as well as non-linear, or linear.

Like MGCFA, moderated nonlinear factor analysis (MNLFA) constitutes a fundamentally confirmatory approach to the assessment of measurement invariance. However, by relaxing several assumptions that are fixed in traditional MGCFA models, MNLFA places even greater demands on model specification and therefore requires more explicit modeling decisions by the researcher. In practice, at least three sets of decisions are required. First, the measurement model must be specified, including the latent structure and the relationships between latent and observed variables. Second, one or more moderator variables must be selected to represent potential sources of heterogeneity in the measurement process. Third, moderation functions must be specified for all parameters that are allowed to vary as a function of the moderator. This includes determining which parameters are moderated as well as specifying the functional form of each moderation effect (e.g., linear, quadratic, or another parametric form). Consequently, the detection of measurement non-invariance depends not only on the choice of moderator variables but also on the assumptions made about how these moderators influence the model parameters.

Structural Equation Model Trees

Structural Equation Model Trees (SEM Trees) were originally introduced by Brandmaier et al. (2013) as a way to systematically search for important covariates and their interactions in data, creating homogeneous subgroups. The method was refined later on by Brandmaier et al. (2016) and Arnold, Voelkle, and Brandmaier (2021). SEM Trees build on two paradigms: SEMs and on decision trees. Decision trees describe recursive partitions of the covariate space, whereby the sample is repeatedly split according to covariate values into progressively more homogeneous subsets, in order to maximize differences in a given outcome variable. SEM Trees combine the hierarchical structure of the decision trees with SEMs, so that they describe heterogeneity in data with respect to a specified SEM as targeted outcome structure. They differ from other model-based tree approaches, as they specifically allow to extend this general logic to models that encompass latent factors. Thus, following the rationale of MNI from MNLFA testing, it becomes apparent, that SEM trees should inherently be suited to test for MNI in data. This is implicitly also suggested in the study on

SEM trees from Arnold et al. (2021), where an example case is used that concerns MI. Nevertheless SEM trees are rarely if at all used to test for MNI in empirical research.

In contrast to MGCFA and MNLFA, SEM trees represent a more exploratory approach to the assessment of MI. Similar to these methods, SEM trees require the researcher to specify a measurement model a priori. Beyond this, however, the researcher only needs to provide a set of candidate partitioning variables that may account for heterogeneity in the data. These variables may be numerous and can be either categorical or continuous. In this respect, the researcher decisions required by SEM trees closely resemble those of the traditional MGCFA framework.

In both cases, a measurement model is specified and variables that may induce heterogeneity are considered. The crucial difference is that MGCFA requires the researcher to select a specific grouping variable and define its manifestation in advance, whereas SEM trees can evaluate a large set of candidate variables simultaneously and determine which, if any, are associated with violations of MI. Moreover, SEM trees are able to identify nonlinear forms of moderation and data-driven partition points for continuous variables without requiring these relationships to be specified a priori.

Given the specified measurement model, the SEM tree algorithm recursively partitions the sample to identify subgroups that differ with respect to model parameters. Consequently, SEM trees retain the confirmatory component associated with the measurement model while adopting an exploratory approach to the identification of heterogeneity. This combination makes them particularly attractive for researchers who are confident in their measurement model but have limited theoretical guidance regarding the sources or structure of MNI. Because the required researcher decisions largely mirror those of the familiar MGCFA framework, the transition from traditional MI testing to SEM trees should involve a comparatively low methodological threshold. At the same time, SEM trees offer substantially greater flexibility by accommodating multiple candidate moderators, continuous and categorical partitioning variables, and potentially nonlinear forms of heterogeneity. Although the explicit specification of moderation mechanisms, as required, for example, in MNLFA, remains advantageous when strong theoretical expectations are available, we believe SEM trees provide a valuable alternative for exploratory analyses and early-stage research in which the relevant sources of heterogeneity are not yet well understood.

Aim of Study

The aim of this simulation study is to evaluate different methods (i.e. MNLFA and SEM trees) regarding their accuracy, power, and false-positive rate in detecting MNI across different factor models. To enhance the study's relevance for empirical practice, the simulation is grounded in a realistic applied-research perspective. Specifically, we consider the situation of a researcher whose understanding of measurement invariance testing is rooted in the traditional MGCFA framework and who seeks to extend this approach to more flexible methods for detecting heterogeneity. Accordingly, MNLFA and SEM trees are evaluated not only as statistical procedures but also as methodological alternatives that researchers may adopt when moving beyond conventional MGCFA-based invariance testing.

Method

Different population SEMs with latent variables and continuous indicators will be simulated in two different simulation studies. The simulation is split into two studies, to separate them on the broad structure of the population SEMs. In study 1 the population model (and its variations) always contains one latent factor, whereas in study two the population model always encompasses two latent factors. Study 1 focuses on functional form misspecifications and data-related conditions. Study 2 builds on Study 1 and introduces additional model misspecifications in the factor structure.

Core Methodological Framework

The data will be generated parametrically. The simulation design includes several manipulated factors that govern the data-generating mechanism (DGM). These factors are varied across simulation conditions and applied consistently to both studies, unless otherwise noted. The following factors are systematically varied in both studies using a grid search design:

Functional Form of Moderation

The effective moderator $Z = h(M)$ is evaluated systematically using the following DGM functions:

1. Linear
2. Quadratic
3. Sigmoid
4. Noise/null moderation, $h(M) = 0$

Strength of Moderation

In each case where parameters are subjected to moderation, the strength of the moderation δ is varied:

δ_λ is evaluated at the following discrete levels:

$$\delta_\lambda \in \{-0.3,\, -0.2,\, 0.2,\, 0.3\}.$$

$$\delta_\nu \in \{-1.0,\, -0.5,\, 0.5,\, 1.0\} \quad \text{for } i \in \{1,2\}, \qquad \delta_\nu = 0 \quad \text{for } i \in \{3,4\},$$

Model Misspecification of Moderator Form

The DGM always defines the effective moderator through $Z = h(M)$. In MNLFA, the analytical model is fitted using prespecified parametric forms, namely linear and quadratic moderation. Consequently, MNLFA is correctly specified when the fitted analytical form corresponds to the DGM transformation and misspecified otherwise. SEM trees do not impose a parametric functional form, but they must identify the correct moderator variable(s) from the candidate set. The simulation assumes that the moderators are always perfectly measured. So this really reflects an ideal scenario and should be kept in mind, when interpreting the results. We expect that one should see a similar picture, but with less-than-perfect reliabilities and lower power in general, should one add variation to the reliability of the moderators.

Performance Criteria

The primary estimands in this simulation study pertain to the detection and accurate estimation of MNI due to moderation effects on measurement parameters. Specifically, we assess whether the estimation methods compared can correctly identify the presence or absence of moderation in factor loadings, intercepts, and residual variances, reflecting the true data-generating structure. Consequently, key simulation outcomes include Type I error rates and statistical power associated with detecting such moderation effects, as well as bias and variability in the recovered parameter estimates. Secondary estimands include model-level indicators of fit (e.g., Kullback–Leibler divergence, AIC), which serve as auxiliary diagnostics for methods that yield such metrics (i.e., MNLFA). These are not to the same extent applicable to tree-based approaches and are thus interpreted as method-specific targets rather than general evaluative criteria.

For the parametric MNLFA, we will extract estimated moderated parameters (e.g., factor loadings, intercepts), their associated standard errors, and two-sided p-values for testing the null hypothesis of no moderation effect. Additionally, model fit indices such as Kullback–Leibler divergence, AIC and the LRT-statistics will be recorded. For the parameter estimates and fit indices, we will summarize their distributions across simulation replications by reporting key quantiles (2.5th, 50th, and 97.5th percentiles). The null hypothesis will be rejected if the associated p-value is less than the conventional significance level of 0.05. In contrast, non-parametric tree-based approaches do not yield direct parameter estimates with standard errors. We use score-based SEM-trees (Arnold et al. 2021). Their detection of MI/non-invariance is operationalized via the presence of at least one split in the correct condition at the correct respective level of focus parameters. Those splits, introducing a subgroup, will be evaluated at the significance level: 0.05. Additionally, a Bonferroni correction to account for multiple comparisons at each node. The trees will be allowed to split on the focus parameters relevant for the invariance testing (the means of the intercepts, factor loadings and the latent covariance in study 2). Candidate subgroup structures are assessed by testing whether they produce statistically significant differences in the specified model parameters across groups. Differences in factor variances between subgroups are permitted in the model, but they are not used as criteria for selecting or evaluating covariates. Furthermore, pre-pruning will be applied, limiting the maximum depth of the tree to 3 and min.**N** is set to 50. Because the MNLFA also directly considers the form of the moderation, which the SEM trees do not, we further want to use SEM forests for generating partial dependence plots to get an approximation of the functional form of the moderating variables.

Replications

We employ a full grid search, in which parameter values are sampled from pre-specified discrete values. To ensure computational feasibility however, we followed an iterative approach whereby a initial pilot simulations $n = 100$ was conducted to amongst others, estimate the needed simulation time. These estimates then informed the total number of XX replications.

Software and Implementation

All simulations and analyses will be conducted in R (R Core Team 2025). The MNLFA-style models will be implemented directly in `OpenMx`(Boker et al. 2026), using raw-data maximum likelihood estimation and definition-variable-style moderation through OpenMx algebra objects. SEM tree analyses will be performed with the `semtree` (Brandmaier et al. 2026) package using OpenMx RAM models. Simulation design construction, data handling, aggregation, and visualization will rely on `dplyr`(Wickham et al. 2026), `tidyr` (Wickham et al. 2025), `tibble` (Müller and Wickham 2026), `purrr` (Wickham and Henry 2026), `furrr` (Vaughan and Dancho 2022), `future` (Bengtsson 2021), `parallel`

(R Core Team 2025), and `ggplot2` (Wickham 2016). The simulation study will be conducted in R (R Core Team 2025).

Study 1

Population Models

Simulation Conditions

Focuses on functional form misspecification and data-related conditions:

- **Sample sizes:** $N \in \{300, 500, 700, 1,000\}$
- **Intercepts:** Intercept baselines are set to 1 across simulation conditions.
- **Indicator reliabilities:** *Low* (.6), *moderate* (.7), *moderate to high* (.8), or *high* (.95).

Results: Study 1

Study 2

Population Models

Simulation Conditions

Builds on Study 1 and introduces additional model misspecification in the factor structure:

- **Structural model misspecifications:** (between-subjects factor)
  - **1. No misspecification:** Factor structure matches the analysis model.
  - **2. Cross-loadings:** Items y_5, y_6 additionally load on a second latent factor (See Figure 3).
- **Latent Covariance:** The latent covariance is moderated by Z. The baseline variance is set to 1 and moderation strength is selected from:

$$\delta_{\sigma_\eta} \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

Results: Study 2

Discussion

Practical Recommendations

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